

The Exact Asymptotic-Freeness Threshold for $\ell_\nu^\infty(\mu)$

A solution of the final-remarks problem in Talponen's paper

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Abstract

Let $\nu < \mu$ be infinite cardinals and let $\ell_\nu^\infty(\mu)$ denote the Banach space of bounded scalar functions on μ whose supports have cardinality at most ν . Talponen asked for the least uncountable regular cardinal κ for which this space is (λ, κ) -asymptotically free. We first check that the supplied partial result, giving the sufficient condition $2^\nu < \kappa$, is correct. We then prove the exact answer: among uncountable regular κ with $\lambda < \kappa$, the space is (λ, κ) -asymptotically free if and only if $\nu < \kappa$. Consequently,

$$\kappa(\lambda, \nu, \mu) = (\max\{\lambda, \nu\})^+.$$

The proof uses only Hahn–Banach and a sharp tail-space counterexample. In particular, the answer is independent of μ , and the hypothesis $\lambda < 2^\nu$ is not needed for the positive implication.

1 The problem

We work over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Following Talponen [1], let κ be an uncountable regular cardinal and $\lambda < \kappa$. A Banach space X is (λ, κ) -asymptotically free if, whenever $Z \subset X$ is a closed subspace with $\text{dens}(Z) \leq \lambda$ and $(Z_\alpha)_{\alpha < \kappa}$ is a decreasing nested family of closed subspaces satisfying

$$Z^\perp = \bigcup_{\alpha < \kappa} Z_\alpha^\perp,$$

then, for every $x \in X$,

$$\|x + z\| \geq \|z\| \quad (z \in Z)$$

implies that there is $\alpha_0 < \kappa$ such that

$$\|x + z\| \geq \|z\| \quad (z \in Z_\alpha, \alpha_0 < \alpha < \kappa).$$

The space is κ -asymptotically free if this holds for every $\lambda < \kappa$.

For infinite cardinals $\nu < \mu$, put

$$\ell_\nu^\infty(\mu) = \{x \in \ell^\infty(\mu) : |\text{supp}(x)| \leq \nu\},$$

with the supremum norm. Talponen's final remarks ask for the least $\kappa = \kappa(\lambda, \nu, \mu)$ such that this space is (λ, κ) -asymptotically free, in the range $\lambda < 2^\nu$ [1, Section 6].

monotone basic sequence. □

6. FINAL REMARKS

We do not know if the spaces generated by an SD sequence and with Corson's property (C) are essentially WLD in some sense. Also, we do not know, given infinite cardinals $\nu < \mu$, $\lambda < 2^\nu$, what is the least $\kappa = \kappa(\lambda, \nu, \mu)$ such that $\ell_\nu^\infty(\mu)$ is (λ, κ) -a.f. If μ is larger than ν then $\ell_\nu^\infty(\mu)$ is relatively 'spread out'. Thus this space can be viewed as being preconditioned towards a free structure alternative in a hypothetical combinatorial dichotomy. A similar cardinal invariant κ involving all reflexive Banach spaces of density $\geq \kappa$ seems reasonable. In Theorem 4.4 and Corollary 4.7 we could have managed with a Banach lattice ideal $\mathcal{I} = \mathcal{I}_{\lambda, \kappa} \subset \ell^\infty(\mu)$, in place of $\ell_\lambda^\infty(\mu)$, such that for any subspace $Y \subset \mathcal{I}$, $\text{dens}(Y) \leq \lambda$, the projection band ideal $J \subset \ell^\infty(\mu)$, generated by Y and with band projection $P_J: X \rightarrow J$, satisfies $\text{dens}(P_J(\mathcal{I})) < \kappa$. Corollary 4.7 suggests an interesting problem regarding a possible *gap* in the cardinalities of unconditional basic sequences: Is it true that each reflexive Banach space of density $\mathfrak{nc}_{\text{rf}}$ already admits an unconditional basic sequence of length ω_1 ?

The class of Banach spaces with (σ) is closed in a sense.

Proposition 6.1. *Suppose that a Banach space X is (σ) -generated, i.e. there is*

2 Idea of the proof

The earlier upper-bound packet stabilized the whole coordinate band generated by a dense set in Z , which naturally led to the sufficient but too-large threshold $2^\nu < \kappa$. The exact proof avoids stabilizing Z at all. For a fixed test vector x , only the coordinates in $\text{supp}(x)$ matter. If x is already monotone over Z , a Hahn–Banach correction produces, for each coordinate $\gamma \in \text{supp}(x)$, an annihilator of Z that forces the γ -coordinate of every sufficiently late Z_α -vector to be controlled by $\|x + y\|$. Since $|\text{supp}(x)| \leq \nu$, regularity of κ synchronizes all these coordinates exactly when $\nu < \kappa$.

The lower bound is equally local. When $\kappa \leq \nu$, place a copy of $c_0(\kappa)$ inside $\ell_\nu^\infty(\mu)$, take the decreasing tail subspaces, and test against the vector that is constantly $-1/2$ on κ . Every functional on the ambient space eventually annihilates the tail, but every tail still contains a unit vector whose addition drops the norm to $1/2$. This gives the sharp obstruction $\kappa \leq \nu$.

3 Assessment of the supplied partial result

The supplied packet proves that $2^\nu < \kappa$ is sufficient. That result is correct. More precisely:

Proposition 1. *Let $\nu < \mu$ be infinite cardinals, let $\lambda < 2^\nu$, and let κ be uncountable regular with $2^\nu < \kappa$. Then $\ell_\nu^\infty(\mu)$ is (λ, κ) -asymptotically free.*

Verification. The density estimate in the packet is valid: if $|A| < 2^\nu$, then bounded rational-valued functions on supports of size at most ν form a dense family in $\ell_\nu^\infty(A)$ of cardinality at most

$$|[A]^{\leq \nu}| \aleph_0^\nu \leq |A|^\nu 2^\nu \leq (2^\nu)^\nu = 2^\nu.$$

The projection-stabilization lemma is also valid in the form used in the packet, with $E_\alpha = \overline{PZ_\alpha}$ and $E = \overline{PZ}$. The identity $E^\perp = \bigcup_{\alpha < \kappa} E_\alpha^\perp$ uses the implicit fact that $Z \subset Z_\alpha$ for every α , which follows from $Z_\alpha^\perp \subset Z^\perp$. The quotient-sphere argument then forces $E_\beta = E$ at some stage $\beta < \kappa$. Applying this to the coordinate band generated by x and a dense subset of Z gives the claimed tail inequality. Thus the partial proof is sound, although its cardinal bound is not optimal. □

4 A support-synchronization lemma

The key point is that one does not need to stabilize the whole coordinate band generated by Z . Only the support of the single test vector x matters.

Lemma 2 (Coordinate correction). *Let X be a linear subspace of $\ell^\infty(\Gamma)$, equipped with the supremum norm. Let $Z \subset X$ be a closed subspace and let $x \in X$ satisfy*

$$\|x + z\| \geq \|z\| \quad (z \in Z).$$

For every $\gamma \in \Gamma$ there is a functional $u_\gamma \in X^$ such that*

$$\|u_\gamma\| \leq 1, \quad u_\gamma(x) = 0, \quad u_\gamma(z) = z(\gamma) \quad (z \in Z).$$

Consequently, if δ_γ denotes coordinate evaluation and $g_\gamma = \delta_\gamma - u_\gamma$, then $g_\gamma \in Z^\perp$, and

$$g_\gamma(y) = 0 \implies |y(\gamma)| \leq \|x + y\|.$$

Proof. If $x = 0$, take $u_\gamma = \delta_\gamma|_X$. Suppose that $x \neq 0$. Then $x \notin Z$, since otherwise the choice $z = -x$ would give $0 \geq \|x\|$.

On the direct sum $W = Z \oplus \mathbb{F}x$, define

$$Q(z + tx) = z.$$

The hypothesis implies that Q is contractive. Indeed, for $t \neq 0$,

$$\|z + tx\| = |t| \|x + z/t\| \geq |t| \|z/t\| = \|z\|,$$

and the case $t = 0$ is immediate. Hence

$$v_\gamma(z + tx) = z(\gamma) = \delta_\gamma(Q(z + tx))$$

defines a functional on W of norm at most one. By Hahn–Banach it extends to a functional $u_\gamma \in X^*$ of norm at most one. It has the stated values on Z and on x .

Now $g_\gamma = \delta_\gamma - u_\gamma$ annihilates Z . If $g_\gamma(y) = 0$, then

$$|y(\gamma)| = |u_\gamma(y)| = |u_\gamma(x + y)| \leq \|x + y\|.$$

□

Theorem 3 (Positive direction). *Let κ be uncountable regular, and let $X \subset \ell^\infty(\Gamma)$ be a Banach space with the inherited supremum norm. If every $x \in X$ has $|\text{supp}(x)| < \kappa$, then X is κ -asymptotically free.*

Proof. Fix $\lambda < \kappa$, a closed subspace $Z \subset X$, a decreasing family $(Z_\alpha)_{\alpha < \kappa}$ satisfying $Z^\perp = \bigcup_{\alpha < \kappa} Z_\alpha^\perp$, and $x \in X$ such that $\|x + z\| \geq \|z\|$ for all $z \in Z$. Put $S = \text{supp}(x)$, so $|S| < \kappa$.

For each $\gamma \in S$, apply Lemma 2 and put $g_\gamma = \delta_\gamma - u_\gamma \in Z^\perp$. Choose $\alpha_\gamma < \kappa$ with $g_\gamma \in Z_{\alpha_\gamma}^\perp$. Regularity of κ and $|S| < \kappa$ yield

$$\alpha_0 = \sup_{\gamma \in S} (\alpha_\gamma + 1) < \kappa.$$

Let $\alpha \geq \alpha_0$ and $y \in Z_\alpha$. For $\gamma \in S$, nestedness gives $g_\gamma(y) = 0$, and hence

$$|y(\gamma)| \leq \|x + y\|.$$

For $\gamma \notin S$, one has $x(\gamma) = 0$, so

$$|y(\gamma)| = |(x + y)(\gamma)| \leq \|x + y\|.$$

Taking the supremum over Γ gives $\|y\| \leq \|x + y\|$. This is the required tail inequality. Since $\lambda < \kappa$ was arbitrary, X is κ -asymptotically free. $\square$

Corollary 4. *If $\nu < \kappa$ and κ is uncountable regular, then $\ell_\nu^\infty(\mu)$ is κ -asymptotically free, for every cardinal μ .*

Proof. Every vector has support of cardinality at most $\nu < \kappa$, so Theorem 3 applies. $\square$

5 The sharp counterexample

Theorem 5 (Negative direction). *Let $\nu < \mu$ be infinite cardinals. Let κ be an uncountable regular cardinal with $\kappa \leq \nu$, and let $\lambda < \kappa$ be an infinite cardinal. Then $\ell_\nu^\infty(\mu)$ is not (λ, κ) -asymptotically free.*

Proof. Choose a set $\Gamma \subset \mu$ of cardinality κ , identify it with the ordinal κ , and regard

$$Y = c_0(\kappa)$$

as a closed subspace of $\ell_\nu^\infty(\mu)$, by extending functions by zero outside Γ . This inclusion is legitimate because every element of $c_0(\kappa)$ has countable support, hence support of size at most ν .

Set $Z = \{0\}$, and for $\alpha < \kappa$ put

$$Z_\alpha = c_0([\alpha, \kappa)) = \overline{\text{span}}\{e_\xi : \alpha \leq \xi < \kappa\}.$$

These are decreasing closed subspaces. We claim that

$$X^* = Z^\perp = \bigcup_{\alpha < \kappa} Z_\alpha^\perp, \quad X = \ell_\nu^\infty(\mu).$$

Indeed, for $f \in X^*$, the restriction $f|_Y$ belongs to $Y^* = \ell^1(\kappa)$, so its support is countable. Since κ is uncountable regular, that support is bounded in κ . Hence $f|_{Z_\alpha} = 0$ for some $\alpha < \kappa$, which proves the claim.

Finally, define $x \in X$ by

$$x(\xi) = -\frac{1}{2} \quad (\xi < \kappa), \quad x(\eta) = 0 \quad (\eta \in \mu \setminus \Gamma).$$

Its support has cardinality $\kappa \leq \nu$. The hypothesis on $Z = \{0\}$ is vacuous. But for every $\alpha < \kappa$, choosing any $\xi \in [\alpha, \kappa)$ gives $e_\xi \in Z_\alpha$ and

$$\|e_\xi\| = 1, \quad \|x + e_\xi\| = \frac{1}{2}.$$

Thus the required inequality fails at every stage of the family, and the space is not (λ, κ) -asymptotically free. $\square$

6 Exact value of the invariant

Theorem 6 (Exact answer). *Let $\nu < \mu$ and λ be infinite cardinals. For every uncountable regular cardinal κ with $\lambda < \kappa$,*

$$\ell_\nu^\infty(\mu) \text{ is } (\lambda, \kappa)\text{-asymptotically free} \iff \nu < \kappa.$$

Consequently, in the notation of Talponen’s final-remarks problem,

$$\boxed{\kappa(\lambda, \nu, \mu) = (\max\{\lambda, \nu\})^+}.$$

In particular, this holds throughout the originally requested range $\lambda < 2^\nu$.

Proof. The forward implication is Theorem 5, and the reverse implication is Corollary 4. The least uncountable regular cardinal larger than both λ and ν is the successor cardinal $(\max\{\lambda, \nu\})^+$. $\square$

Remark 7. The positive proof uses neither $\text{dens}(Z) \leq \lambda$ nor $\lambda < 2^\nu$, beyond the formal requirement $\lambda < \kappa$ in the definition. It also shows that, for every uncountable regular κ , the Banach space $\ell_{<\kappa}^\infty(\mu)$ is κ -asymptotically free whenever it is complete. Thus the same argument strengthens the strongly-inaccessible hypothesis appearing in Corollary 4.5 of [1].

7 Literature check

A targeted search was made for the exact paper title, arXiv:1604.04408, the phrases “asymptotically free Banach space” and “ (λ, κ) -asymptotically free”, and the specific notation $\ell_\nu^\infty(\mu)$, together with citation-oriented searches for Talponen’s paper. The local run indexes and ledgers were also checked for arXiv:1604.04408, Talponen, asymptotic freeness, and the $\ell_\nu^\infty(\mu)$ final-remarks problem. The searches located the source paper, the earlier upper-bound partial packet, and indexing or announcement pages, but no later paper giving either the exact invariant above or a counterexample resolving the final-remarks question. This is a targeted rather than logically exhaustive bibliographic search, so the claim here is only that no prior resolution was found.

References

- [1] J. Talponen, *Unconditional and bimonotone structures in high density Banach spaces*, arXiv:1604.04408v2, 2016.